

From two columns to a testable chart choice

Same task. Actual pixels. One score that determines the ranking.

1 RAW OBSERVATIONS

A: benign B: malignant

0.2699	1.0950
0.1852	0.5435
0.2773	0.7456
0.1839	0.4956
0.1563	0.7572



2 RENDER AND RECOVER

21 settings • 7 eligible chart families

Pixels + calibrated axes → five differences

$$S = 100 / (1 + \text{normalized recovery loss})$$



3 SELECT

Heatmap

97.981 / 100

Best under this protocol

All 357 + 212 records analysed

Actual selected raster • reference resolution

A

B



-0.36 1.5 3.3 -0.36 1.5 3.3

Value • White → blue: bin probability 0 → 1

RANK / CHART AND SETTING

SCORE / 100

1	Heatmap 64 bins		97.981
2	Dot plot raw observations		96.956
3	Histogram 64 bins		96.686
4	Violin plot 32 grid points		92.868
5	Bar chart means		60.525
6	Interval plot five-number		52.309
7	Box plot five-number		52.136

Higher is better • Green: computational optimum

15-family catalogue screened; seven implemented for this task. Scores are computational, not human accuracy.